



Robot and Smart Card Operating System

Secure, Card-Authenticated Robotic Control powered by a Lightweight Smart Card Operating System



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Introduction

Robotics and smart-card security are two mature technologies that rarely meet. This project fuses them: a mobile robot whose identity, permissions, and task profile are verified through a dedicated Smart Card Operating System (SCOS) before any motion command is accepted.

Instead of relying on passwords or open wireless pairing, the robot reads a contact/contactless smart card through an ISO/IEC 7816 or ISO/IEC 14443 interface. The card's onboard OS authenticates the user and releases the operating profile to the robot controller.

7816

ISO contact-card standard used

<200ms

Target card authentication time

3

Access levels: Guest / Operator / Admin

Problem Statement & Motivation

THE GAP

- 1 Most low-cost robots accept commands from anyone within wireless range — no identity check.
- 2 Password-based access on embedded controllers is weak, hard to rotate, and easily shared.
- 3 Industrial and campus robots need traceable, per-user accountability for safety-critical actions.

THE OPPORTUNITY

- 1 Smart cards already provide tamper-resistant identity in banking, transit, and access control.
- 2 A dedicated Smart Card OS can store keys, PINs, and role profiles far more securely than flash memory on the robot itself.
- 3 Combining both gives a low-cost, swappable, physically-authenticated control layer for robots.

Objectives

1

DESIGN

Design a mobile robot platform controllable via authenticated commands only.

2

SECURE

Implement a Smart Card OS layer that stores keys, PINs and role-based permissions.

3

INTEGRATE

Build a reliable card-to-controller communication and command-mapping pipeline.

4

VALIDATE

Test authentication speed, access-control correctness, and motion accuracy.

Understanding Robots

A robot is a programmable electromechanical system capable of sensing its environment, processing information, and performing physical actions autonomously or under remote command.



Mobile Robots

Wheeled/tracked
platforms for navigation &
delivery — used in this
project



Industrial Arms

Fixed manipulators for
assembly, welding, and
pick-and-place



Service Robots

Assist in homes, hospitals,
and retail environments



Autonomous Vehicles

Self-driving cars, drones,
and AGVs for logistics

Understanding Smart Cards

A smart card is a pocket-sized card with an embedded microprocessor and secure memory that can compute, store keys, and run a small operating system — not just hold static data like a magnetic stripe.



Contact Cards

ISO/IEC 7816

- Gold contact-pad interface
- Inserted into a card reader
- Used for SIMs, banking EMV
- Reliable, shielded electrical link



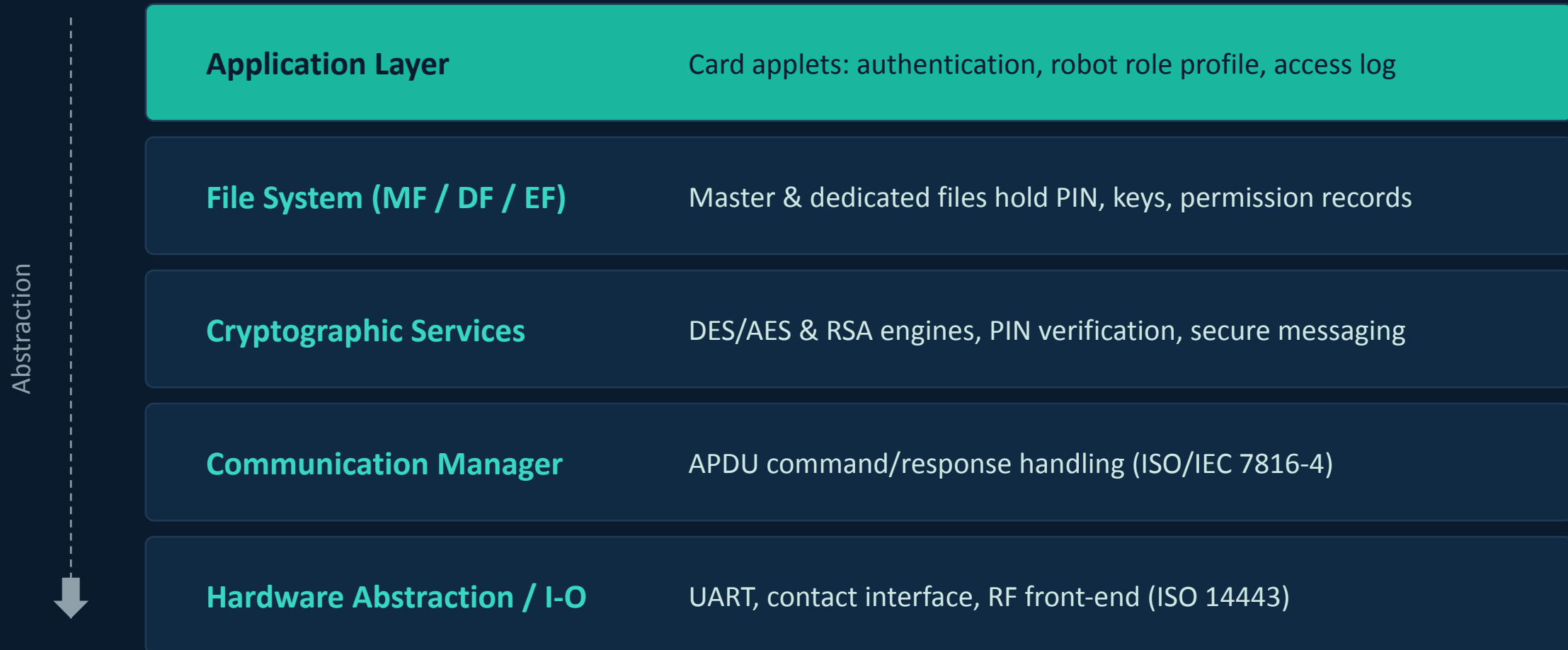
Contactless Cards

ISO/IEC 14443

- RFID / NFC based (13.56 MHz)
- Tap-and-go, no insertion
- Used for transit, access badges
- Faster user interaction

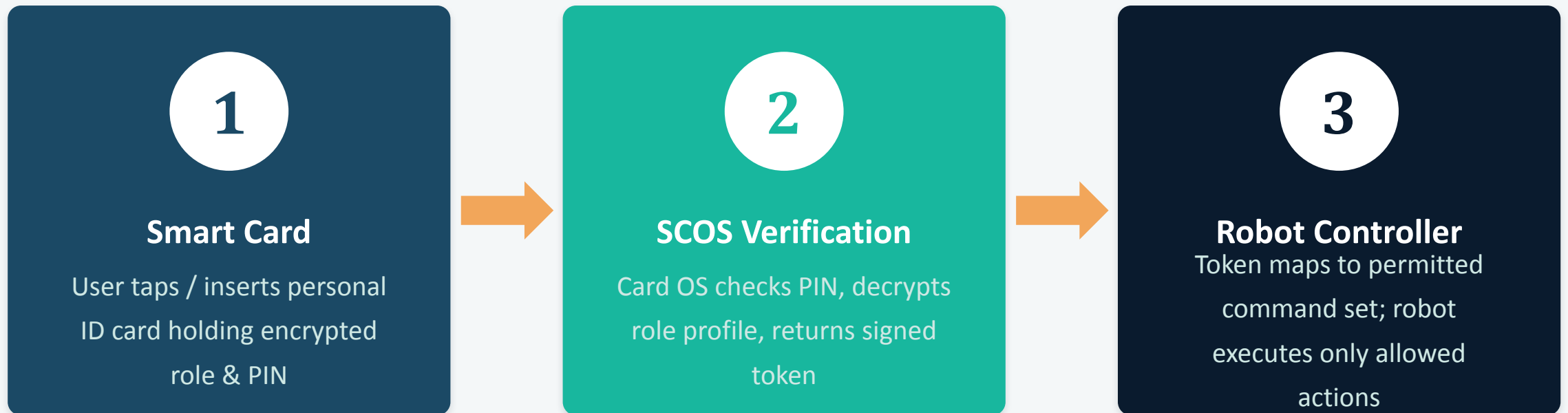
Smart Card Operating System (SCOS)

A minimal, secure OS resident in the card's microcontroller, exposing services through APDU commands.

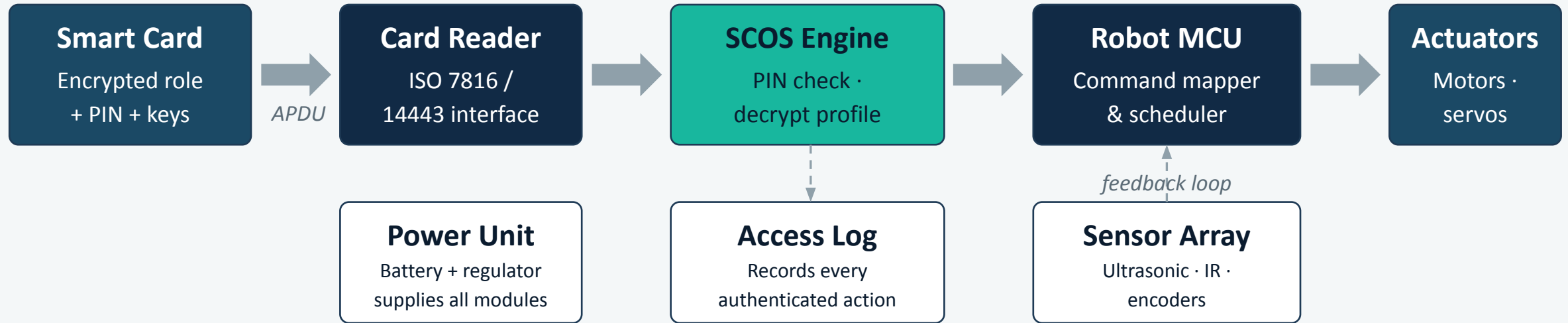


Proposed System Concept

We propose a robot that only executes commands after a smart card presents valid credentials to its onboard SCOS. The card doesn't just unlock the robot — it carries the user's exact permission profile.



System Architecture



Optional: Wi-Fi / Bluetooth telemetry link to monitoring dashboard

Design principle: the robot's actuators never receive a raw command — every instruction is first validated against the role and permission set decrypted from the smart card by the SCOS engine.

Hardware Components



Microcontroller

ATmega328 / STM32 core running
command mapper firmware



Smart Card Module

ISO 7816 contact reader + PN532
NFC front-end



Motor Driver

Dual H-bridge driving two DC gear
motors



Chassis & Wheels

Two-wheel differential drive with
caster support



Sensor Suite

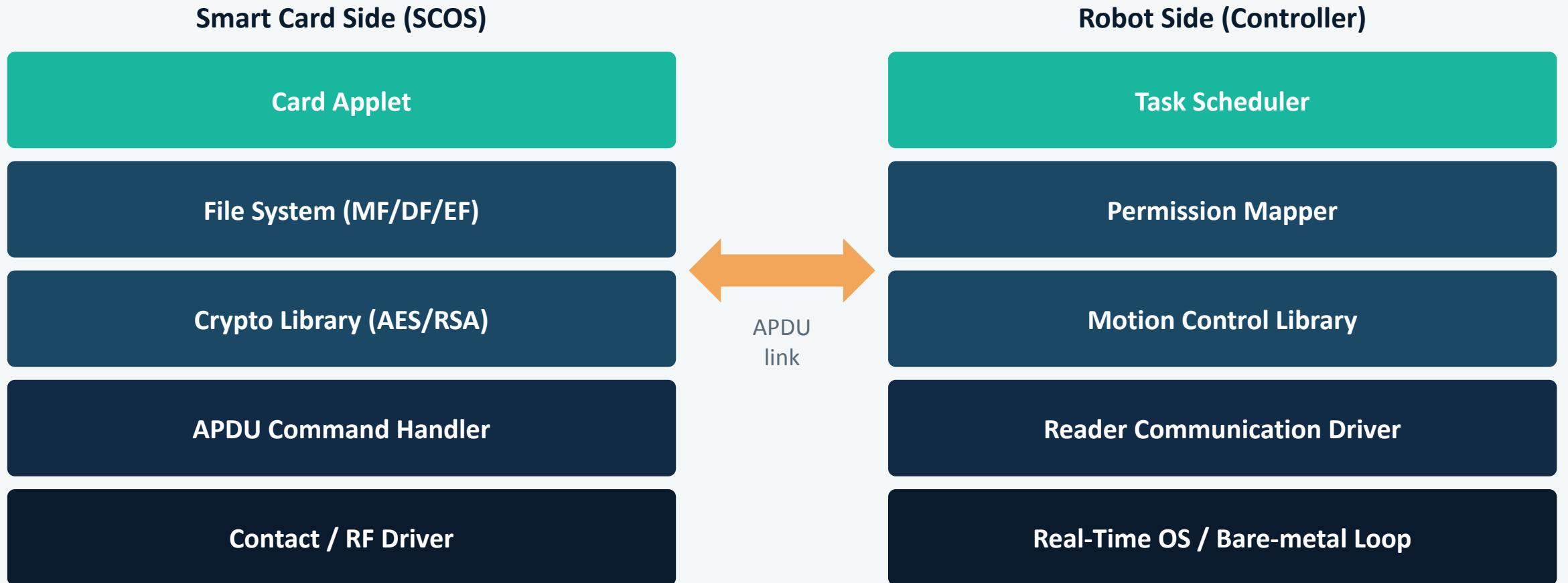
Ultrasonic distance, IR line sensors,
wheel encoders



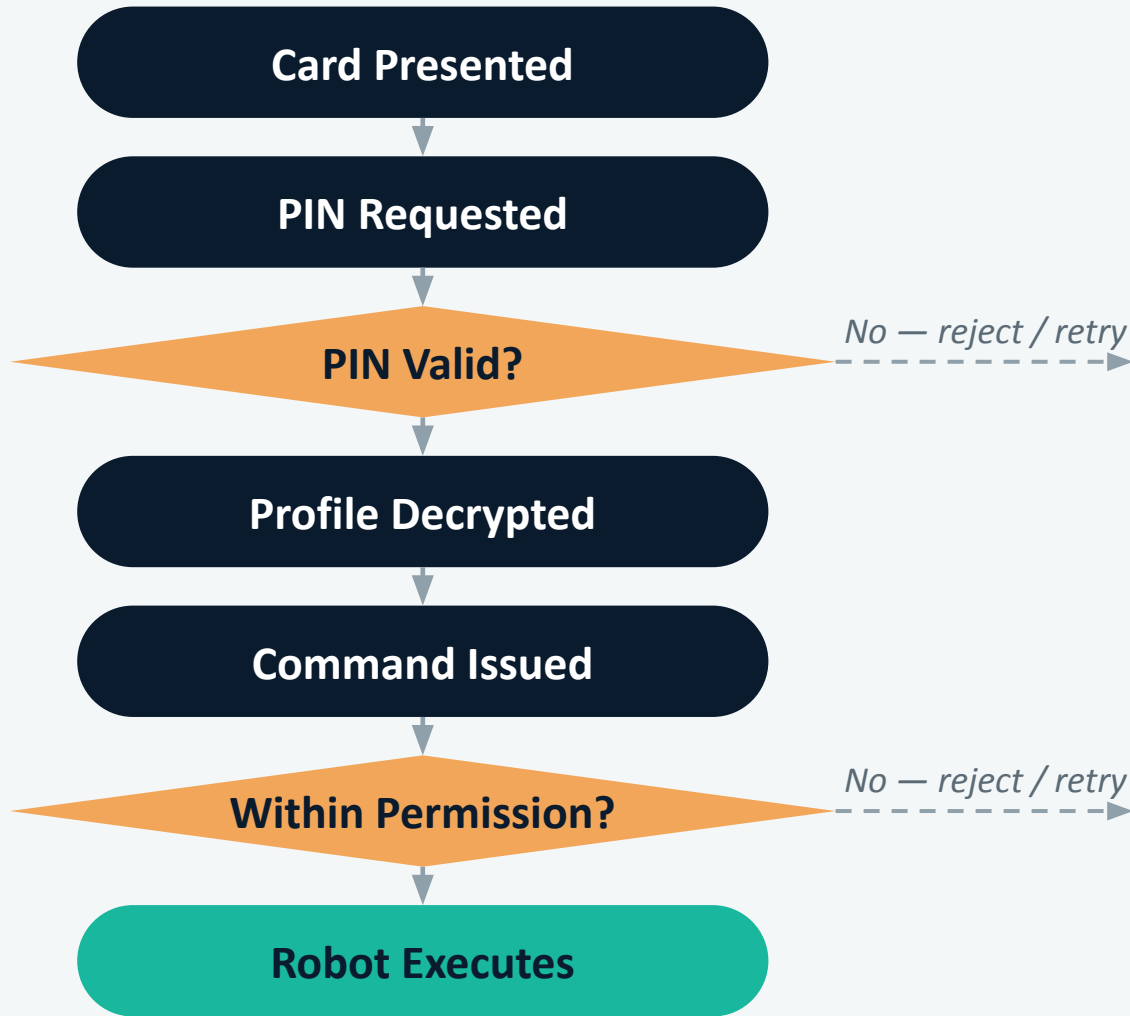
Power Module

Li-ion pack with buck regulator for
logic rails

Software Stack



Working Methodology



Card Presented

User taps/inserts smart card at the robot's reader module.

PIN Requested

SCOS challenges with a PIN prompt via keypad or companion app.

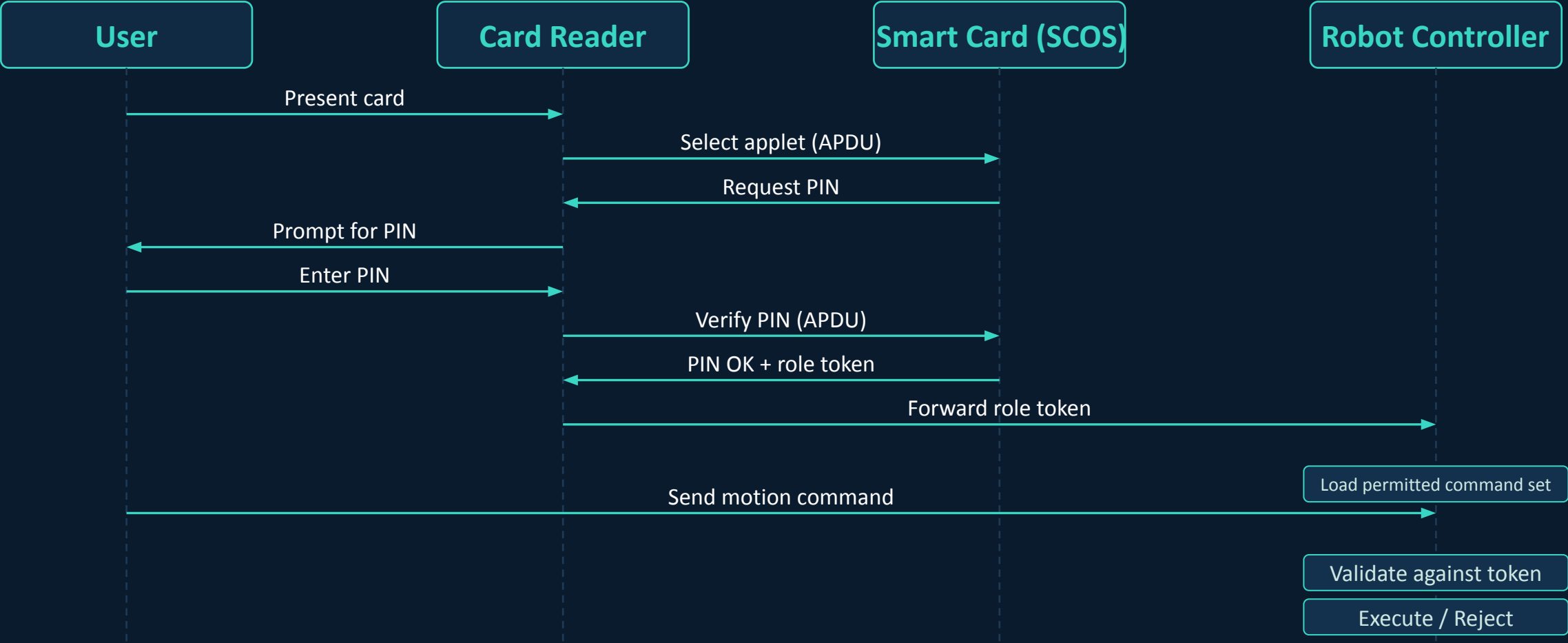
Profile Decrypted

On success, role & permitted command list are released.

Robot Executes

Command runs only if it matches the decrypted permission set.

Authentication Sequence



Command Execution & Access Control

Every command the robot receives is checked against the role decrypted from the card. The matrix below defines exactly what each role may execute.

Command	Guest	Operator	Admin
Move Forward / Reverse			
Turn Left / Right			
Adjust Speed Profile			
Access Sensor Data Log			
Reprogram Task Sequence			
Update Card Permissions			
Firmware / OTA Update			

Applications

1

Warehouse & Logistics

Only certified operators dispatch
AGVs carrying inventory

2

Hospitals

Nurses tap ID cards to run
medicine-delivery robots on
approved routes

3

Campus & Office Security

Patrol robots activated solely by
security staff credentials

4

Manufacturing Floors

Role-based control of robotic arms
prevents unauthorized
reprogramming

5

Smart Homes

Family members get tiered control
over home assistant robots

6

Defense & Restricted Zones

High-assurance card + PIN gate for
sensitive robotic operations

Advantages



Physical + digital factor: losing a password is easy, cloning a smart card's crypto keys is not.



Every action is attributable to the card holder — strong audit trail for safety incidents.



Role separation keeps guests, operators, and admins on clearly bounded command sets.



Cards are swappable and reusable across many robots sharing the same SCOS standard.



Card-side logic keeps the robot's firmware simple, cheap, and easy to update.



Standards-based (ISO 7816 / 14443) design interoperates with existing card infrastructure.

Challenges & Limitations



Latency

APDU round-trips add tens of milliseconds — real-time control loops must budget for it.



Environmental Exposure

Outdoor/industrial robots need ruggedized readers resistant to dust and vibration.



Card Loss / Theft

A stolen but un-PIN-protected card could still grant guest-level access.



Cost per Unit

Smart card + certified reader adds bill-of-materials cost versus a simple keypad.



Limited On-Card Memory

Complex permission sets are constrained by the card's EEPROM size.

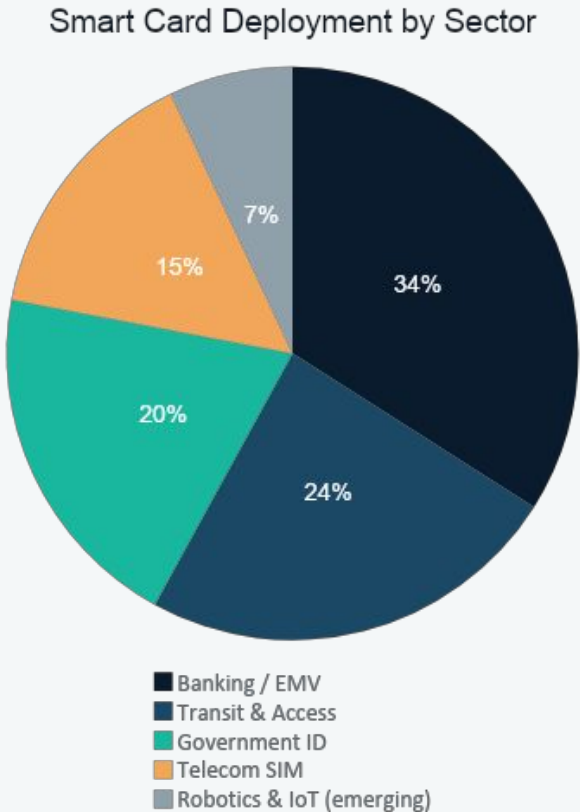
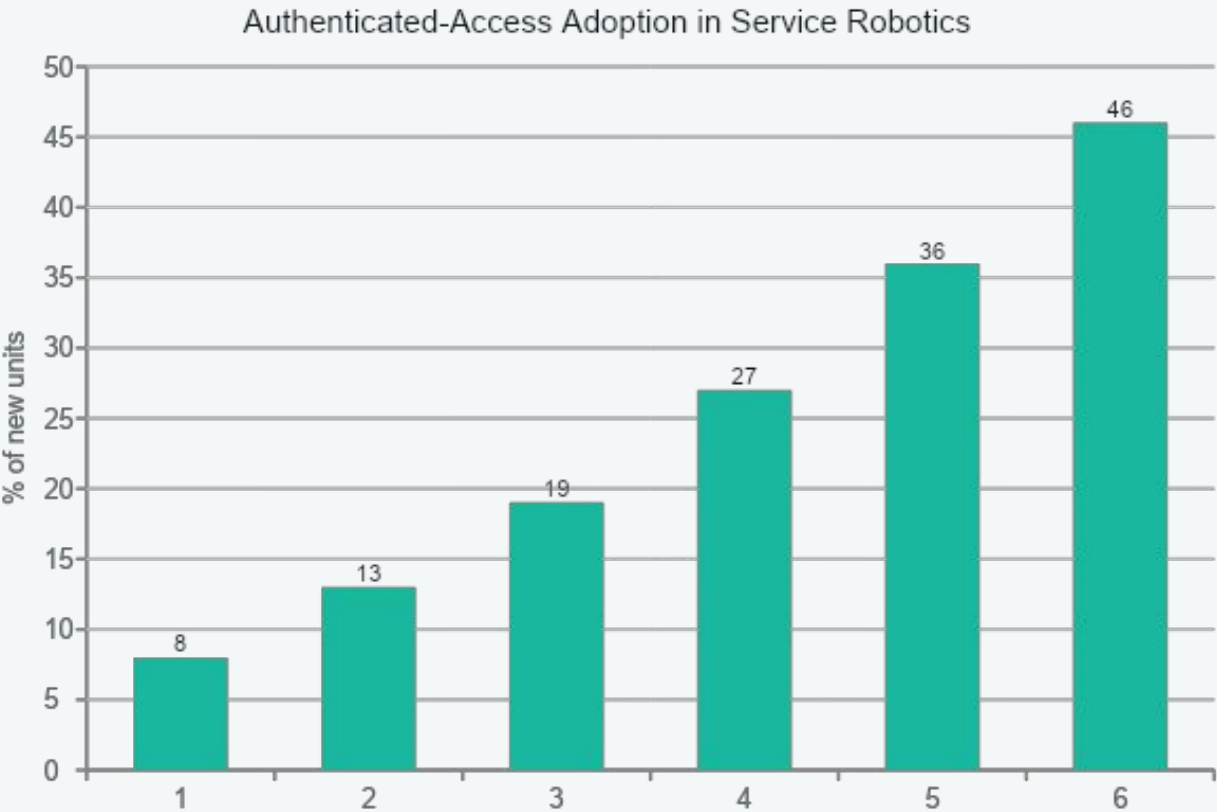


Firmware Update Complexity

Card applet updates require a secure over-the-air or physical re-issue process.

Market Outlook

Figures below are illustrative projections used to motivate the project's relevance, not audited market data.



Traditional vs Smart-Card-Enabled Robots

Criteria	Traditional Robot	Smart-Card-Enabled Robot
Access Control	Open pairing or shared password	Per-user card + PIN, role-scoped
Auditability	Little to no action trace	Every command logged to a card ID
Credential Revocation	Requires firmware/password change	Instantly disable a single card
Onboarding New Users	Manual reconfiguration	Issue a pre-provisioned card
Tamper Resistance	Software-only, easily bypassed	Hardware secure element on card
Hardware Cost Impact	Lower	Slightly higher (reader + card)

Future Scope



Biometric Fusion

Combine fingerprint or face match with the card for two-factor robot access



Cloud Fleet Management

Sync SCOS access logs across a fleet dashboard for central audit



Dynamic Role Updates

Push temporary permission changes to a card over a secure OTA channel



AI-Assisted Anomaly Detection

Flag unusual authenticated command patterns as possible misuse



Multi-Robot Card Federation

One card securely operating an entire authorized robot fleet



Miniaturized Secure Element

Move SCOS logic into a wearable token for hands-free authentication

Conclusion

This project shows that pairing a mobile robot with a Smart Card Operating System is a practical, low-cost way to bring hardware-grade identity and role-based access control into robotics — closing a gap that pure software security cannot.

- 1 SCOS provides tamper-resistant, portable identity for robot operators
- 2 Role-based permission mapping keeps guest / operator / admin actions clearly separated
- 3 Standards-based design (ISO 7816 / 14443) keeps the system interoperable and extensible
- 4 The approach scales from a single classroom robot to fleet-wide industrial deployment

References

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- Craig, K. — "Introduction to Robotics: Mechanics and Control," Pearson.
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Thank You

Questions & Discussion Welcome

